

Option B – Medical Physics

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)	Heater to produce / boil off [free] electrons / thermionic emission (1) accept heater gives energy to electrons Avoid collisions [on way to target] (1)	2			2		
		(ii)	$\frac{100 \times 10^{-3}}{1.6 \times 10^{-19}}$ (1) = 6.25×10^{17} or 6.3×10^{17} with no units (1) accept electrons per second		2		2	2	
		(iii)	Application of conservation of energy: $\frac{1}{2} mv^2 = eV$ (1) Rearrangement: $v^2 = \frac{2eV}{m}$ (1) $v = 8.3$ or 8.4×10^7 [m s ⁻¹] (1)		1 1 1		3	3	
		(iv)	Use of $eV = \frac{hc}{\lambda_{\min}}$ (1) Rearrangement: $\lambda_{\min} = \frac{hc}{20000e}$ (1) $\lambda_{\min} = 6.2 \times 10^{-11}$ [m] (1)	1	1 1		3	3	
	(b)	(i)	Alternating voltage used (1) accept alternating current used Causes piezoelectric crystal to vibrate (1) Ultrasound reflected and Doppler shift detected (1)	3			3		